1. Clinical Question & Core Hypothesis

Clinical Question

IDC-P is a morphologic marker of aggressive prostate cancer, but not all IDC-P tumors respond to AR-targeted therapy — why does IDC-P sometimes denote AR-dependent biology and sometimes AR-independent, therapy■resistant disease? 2) Tier 1 paths used: (a) Node1→Node2 (IDC■P correlates with elevated AR transcriptional activity in a subset), and (b) Node2→Node3 (high AR activity predicts response to AR inhibitors). I built the hypothesis on the mechanistic idea that IDC■P tumors are heterogeneous for AR dependence and that AR activity determines sensitivity to AR pathway blockade. 3) Tier 2/supporting evidence: published observations that IDC■P associates with aggressive molecular features but can co■occur with strong AR signaling; neoadjuvant AR inhibitor studies showing deeper pathologic/PSA responses in tumors with high AR■signature scores; and biologic rationale that tumors reliant on AR transcriptional programs are vulnerable to enzalutamide/abiraterone. 4) Endpoint choice: therapy_response (pathologic downstaging, PSA decline, and downstream recurrence/metastasis outcomes) is the most direct, actionable clinical readout for selecting patients for neoadjuvant/adjuvant AR■targeted trials. 5) Exclusions: I deliberately excluded detailed genomic resistance drivers (e.g., BRCA2, PTEN loss, AR splice variants) from the core claim to keep the hypothesis focused on the morphologic–transcriptional interaction; these are listed as secondary modifiers for follow■up, not in the main hypothesis?

Working Hypothesis

IDC-P marks an aggressive tumor state that is heterogeneous for AR dependency. When IDC-P co-occurs with high AR transcriptional activity (AR-high), the tumor remains AR-driven and is susceptible to AR-targeted therapy, whereas IDC-P with low AR activity (AR-low) is AR-independent and less responsive. Notes: genomic resistance determinants are treated as secondary modifiers and excluded from the core hypothesis to maintain focus on the morphologic–transcriptional interaction. Predictions: AR-high IDC-P tumors will show higher AR-signature scores in pretreatment samples and exhibit greater pathologic downstaging and PSA declines in response to AR inhibitors (e.g., enzalutamide, abiraterone) with improved short-term recurrence metrics; AR-low IDC-P tumors will have attenuated responses and may rely on alternative resistance pathways such as lineage plasticity or neuroendocrine features. Mechanistic framing: AR transcriptional programs sustain IDC-P–driven growth only when AR activity is high; AR blockade disrupts these programs in AR-high tumors, whereas AR-low tumors depend on AR-independent pathways that limit responsiveness to AR-targeted therapy.

Element	Value
Imaging feature	IDC■P presence (intraductal carcinoma of the prostate) on diagnostic/surgical specimen
Anchor gene(s)	AR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AR: arker of aggressive prostate cancer, but not all IDC-P tumors respond to AR-targeted therapy — why does IDC-P sometimes denote AR-dependent biology and sometimes AR-independent, therapy■resistant disease? 2) Tier 1 paths used: (a) Node1→Node2 (IDC■P correlates with elevated AR transcriptional activity in a subset), and (b) Node2→Node3 (high AR activity predicts response to AR inhibitors).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Canonical AR target genes (e.g., KLK3, TMPRSS2) comprising the AR transcriptional signature	Supporting modifier from knowledge graph
Secondary	Genomic resistance/ aggression modifiers to examine in stratified analyses (e.g., TP53 loss/mutation, PTEN loss, AR splice variants)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Clinical correlation: assemble IDC-P–positive cohorts treated with AR inhibitors; quantify AR activity in diagnostic specimens using a validated AR transcriptional signature (e.g., a composite score from canonical AR targets such as KLK3, TMPRSS2, NKX3-

Prediction 2: and categorize as AR-high vs AR-low; correlate with outcomes including pathologic response (pCR/downstaging), PSA decline, and recurrence-free survival.

Prediction 3: Preclinical models: develop IDC-P–like organoids/cell lines with variable AR activity; manipulate AR signaling (overexpression/knockdown) and assess sensitivity to enzalutamide/abiraterone; perform RNA-seq and AR ChIP-seq to map AR-dependent transcriptional programs.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- Canonical AR target genes (e.g., KLK3, TMPRSS2) comprising the AR transcriptional signature: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Genomic resistance/ aggression modifiers to examine in stratified analyses (e.g., TP53 loss/mutation, PTEN loss, AR splice variants): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.